TroubleShooting en K8S

CAS TRAINING

```bash
curl http://web-service-ip:node-port
curl: (7) Failed to connect to web-service-ip port node-port: Connection timed out
```

```bash
kubectl describe service web-service
```

```
Name:                     web-service
Namespace:                default
Labels:                   <none>
Annotations:              <none>
Selector:                 name=webapp-mysql
Type:                     NodePort
IP:                       10.96.0.156
Port:                     <unset> 8080/TCP
TargetPort:               8080/TCP
NodePort:                 <unset> 31672/TCP
Endpoints:                10.32.0.6:8080
Session Affinity:         None
External Traffic Policy:  Cluster
Events:                   <none>
```

```yaml
apiVersion: v1
kind: Pod
metadata:
  name: webapp-mysql
  labels:
    app: example-app
    name: webapp-mysql
spec:
  containers:
    - name: webapp-mysql
      image: simple-webapp-mysql
      ports:
        - containerPort: 8080
```

```bash
kubectl describe pod web
```

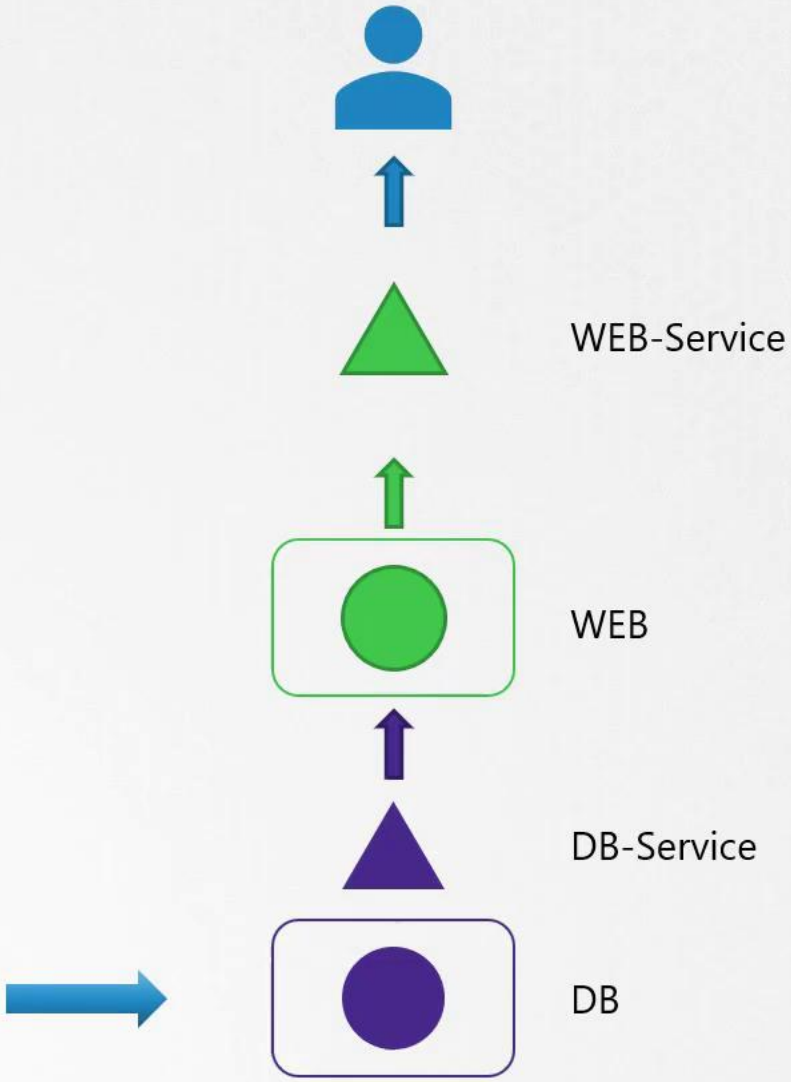
```
...
Events:
Type     Reason     Age   From                Message
----     ------     ----  ----                -------
Normal   Scheduled  52m   default-scheduler   Successfully assigned webapp-mysql to worker-1
Normal   Pulling    52m   kubelet, worker-1   pulling image "simple-webapp-mysql"
Normal   Pulled     52m   kubelet, worker-1   Successfully pulled image "simple-webapp-mysql"
Normal   Created    52m   kubelet, worker-1   Created container
Normal   Started    52m   kubelet, worker-1   Started container
```

```bash
kubectl logs web
```

```
10.32.0.1 - - [01/Apr/2019 12:51:55] "GET / HTTP/1.1" 200 -
10.32.0.1 - - [01/Apr/2019 12:51:55] "GET /static/img/success.jpg HTTP/1.1" 200 -
10.32.0.1 - - [01/Apr/2019 12:51:55] "GET /favicon.ico HTTP/1.1" 404 -
10.32.0.1 - - [01/Apr/2019 12:51:57] "GET / HTTP/1.1" 200 -
10.32.0.1 - - [01/Apr/2019 12:51:58] "GET / HTTP/1.1" 200 -
10.32.0.1 - - [01/Apr/2019 12:51:58] "GET / HTTP/1.1" 200 -
```

kubectl logs web -f

Check Dependent Applications



kubectl get nodes

```
service kube-apiserver status
● kube-apiserver.service - Kubernetes API Server
   Loaded: loaded (/etc/systemd/system/kube-apiserver.service; enabled; vendor preset: (
   Active: active (running) since Wed 2019-03-20 07:57:25 UTC; 1 weeks 1 days ago
   Docs: https://github.com/kubernetes/kubernetes
 Main PID: 15767 (kube-apiserver)
   Tasks: 13 (limit: 2362)
```

```
service kube-controller-manager status
● kube-controller-manager.service - Kubernetes Controller Manager
   Loaded: loaded (/etc/systemd/system/kube-controller-manager.service; enabled; vendor
   Active: active (running) since Wed 2019-03-20 07:57:25 UTC; 1 weeks 1 days ago
   Docs: https://github.com/kubernetes/kubernetes
 Main PID: 15771 (kube-controller)
   Tasks: 10 (limit: 2362)
```

```
service kube-scheduler status
```

```
kubectl logs kube-apiserver-master -n kube-system
```

```
I0401 13:45:38.190735       1 server.go:703] external host was not specified, using 172.
I0401 13:45:38.194290       1 server.go:145] Version: v1.11.3
I0401 13:45:38.819075       1 plugins.go:158] Loaded 8 mutating admission controller(s) s
NamespaceLifecycle,LimitRanger,ServiceAccount,NodeRestriction,Priority,DefaultTolerationS
...
W0401 13:45:41.381736       1 genericapiserver.go:319] Skipping API scheduling.k8s.io/v1a
```

```
sudo journalctl -u kube-apiserver
```

```
kubectl get nodes
NAME        STATUS     ROLES     AGE   VERSION
worker-1    Ready      <none>    8d    v1.13.0
worker-2    NotReady   <none>    8d    v1.13.0
```

```
kubectl describe node worker-1
```

```
service kubelet status
```

```
● kubelet.service - Kubernetes Kubelet
   Loaded: loaded (/etc/systemd/system/kubelet.service; enabled; vendor preset: enabled)
   Active: active (running) since Wed 2019-03-20 14:22:06 UTC; 1 weeks 1 days ago
   Docs: https://github.com/kubernetes/kubernetes
 Main PID: 1281 (kubelet)
   Tasks: 24 (limit: 1152)
```

```
sudo journalctl -u kubelet
```

```
-- Logs begin at Wed 2019-03-20 05:30:37 UTC, end at Mon 2019-04-01 14:42:42 UTC. --
Mar 20 08:12:59 worker-1 systemd[1]: Started Kubernetes Kubelet.
Mar 20 08:12:59 worker-1 kubelet[18926]: Flag --tls-cert-file has been deprecated, This pa
Mar 20 08:12:59 worker-1 kubelet[18926]: Flag --tls-private-key-file has been deprecated,
Mar 20 08:12:59 worker-1 kubelet[18926]: I0320 08:12:59.915179   18926 flags.go:33] FLAG:
Mar 20 08:12:59 worker-1 kubelet[18926]: I0320 08:12:59.918149   18926 flags.go:33] FLAG:
Mar 20 08:12:59 worker-1 kubelet[18926]: I0320 08:12:59.918339   18926 flags.go:33] FLAG:
Mar 20 08:12:59 worker-1 kubelet[18926]: I0320 08:12:59.918520   18926 flags.go:33] FLAG:
Mar 20 08:12:59 worker-1 kubelet[18926]: I0320 08:12:59.918621   18926 flags.go:33] FLAG:
Mar 20 08:12:59 worker-1 kubelet[18926]: I0320 08:12:59.918740   18926 flags.go:33] FLAG:
Mar 20 08:12:59 worker-1 kubelet[18926]: I0320 08:12:59.919874   18926 flags.go:33] FLAG:
Mar 20 08:12:59 worker-1 kubelet[18926]: I0320 08:12:59.919929   18926 flags.go:33] FLAG:
Mar 20 08:12:59 worker-1 kubelet[18926]: I0320 08:12:59.919946   18926 flags.go:33] FLAG:
Mar 20 08:12:59 worker-1 kubelet[18926]: I0320 08:12:59.919948   18926 flags.go:33] FLAG:
Mar 20 08:12:59 worker-1 kubelet[18926]: I0320 08:12:59.919950   18926 flags.go:33] FLAG:
Mar 20 08:12:59 worker-1 kubelet[18926]: I0320 08:12:59.919958   18926 flags.go:33] FLAG:
Mar 20 08:12:59 worker-1 kubelet[18926]: I0320 08:12:59.920160   18926 flags.go:33] FLAG:
Mar 20 08:12:59 worker-1 kubelet[18926]: I0320 08:12:59.920172   18926 flags.go:33] FLAG:
```

```
openssl x509 -in /var/lib/kubelet/worker-1.crt -text
```

```
Certificate:
    Data:
        Version: 3 (0x2)
        Serial Number:
            ff:e0:23:9d:fc:78:03:35
    Signature Algorithm: sha256WithRSAEncryption
    Issuer: CN = KUBERNETES-CA
    Validity
        Not Before: Mar 20 08:09:29 2019 GMT
        Not After : Apr 19 08:09:29 2019 GMT
    Subject: CN = system:node:worker-1, O = system:nodes
    Subject Public Key Info:
        Public Key Algorithm: rsaEncryption
        Public-Key: (2048 bit)
        ...
```